Sandun Munasinghe

Chemical Engineer | AI & Machine Learning | Process Modeling & Optimization

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TECHNICAL SKILLS

Chemical Engineering: Process Control | MATLAB/Simulink | Batch Reactor Scale-Up | Root Cause Analysis
Manufacturing & Quality: GMP | ISO Standards | Quality Control | Process Troubleshooting | Continuous Improvement
Programming/Data: Python | SQL | Pandas | NumPy | Spark | SciPy | ETL Pipelines | Statistical Modeling
Machine Learning: PyTorch | TensorFlow | scikit-learn | Regression | Deep Learning | Neural Networks
Generative AI & LLMs: Transformers | NLP | RAG | LangChain | LangGraph | Hugging Face
Backend/Deployment: FastAPI | AWS (Lambda, S3, API Gateway) | Docker | CI/CD | Git

TECHNICAL PROJECTS

PI Temperature Control of a Dual-Heater Thermal System | *MATLAB/Simulink, Arduino | Rutgers University* *Fall 2024*

- Modeled thermal dynamics with energy-balance ODEs and FOPDT approximations; identified process gains (62.5 K/W) and time constants (~130 s) via nonlinear least-squares regression
- Designed PI controllers using IMC tuning with clamping anti-windup, then fine-tuned to minimize overshoot and improve setpoint tracking
- Achieved under 2°C overshoot and under 200 s rise time across cold-start and fan-disturbance tests, using PWM heater control and live sensor feedback

Multi-Agent LLM Conflict Resolution (GitHub) | NLP Project, Stevens Institute of Technology *Spring 2026*

- Built 4 LangGraph orchestration strategies (majority voting, structured debate, hierarchical authority, evidence-weighted consensus) with sentence-transformer deadlock detection and judge-mediated resolution
- Streamed multi-agent debates in real time via SSE on an async FastAPI backend with a Next.js frontend and batch experiment runner
- Tracked 9 behavioral metrics across 32 scenarios; found misinformation agents outperformed factual agents in 88% of cases (p=0.006)

Chemical Engineering Domain Language Model | *Independent Project,* *Fall 2026*

Python, PyTorch, Transformers, NLP, Data Engineering

- Building a GPT-2–style decoder-only transformer from scratch in PyTorch, to be pretrained on a chemical engineering corpus of research papers and technical references
- Engineered automated pipelines to collect 3,000+ arXiv LaTeX source archives and Wikipedia articles, with text extraction, cleaning, quality scoring, and domain-relevance filtering
- Processed 2,300+ research papers into a curated 10M-token training corpus
- Next phase: extending the model into an AI troubleshooting assistant that diagnoses equipment faults, identifies root causes, and recommends corrective actions, supported by deep learning process models for optimization

PROFESSIONAL EXPERIENCE

Graduate Research Assistant at Stevens Institute of Technology (Advisor: Zining Zhu) *ECAI Lab* *Jan 2026 - Present*

LLM Truthfulness and Interpretability Research (GitHub) - Python | PyTorch | Hugging Face | OpenAI API

- Built an LLM evaluation pipeline using 790 TruthfulQA questions and four prompting conditions to investigate adversarially induced changes in truthfulness
- Analyzed behavioral flips in GPT-5.6 Sol and Gemma 3 4B IT using automated truthfulness evaluation and ROUGE metrics
- Developed a logit-lens framework to investigate layer-wise representational divergence and truthful versus untruthful token evidence under adversarial prompting

Operations & Customer Service Associate Manager | United States Postal Service Oct *2018 – Oct 2023*

- Applied demand variability analysis to redesign staffing models across a 200K+ daily package operation, reducing overtime costs by 20%
- Led deployment of automated sorting systems (SDUS): troubleshot integration issues and trained 20+ staff, improving system adoption
- Communicated performance insights and operational decisions to non-technical stakeholders and senior leadership, building cross-functional communication skills critical in enterprise

Process & Quality Engineering Associate | *Reckitt Benckiser.* *Aug 2012 – Oct 2017*

- Supported batch reactor design, scale-up, and optimization for disinfectant production across global brands (Lysol, Harpic, Air Wick)
- Managed quality control from raw materials through final packaging, including batch testing and real-time corrective actions to meet GMP and ISO standards
- Identified and resolved process bottlenecks, increasing throughput by 40% and reducing material waste by 20%
- Partnered with cross-functional teams on continuous improvement and process documentation, reducing downtime by 10% and strengthening operator safety compliance

EDUCATION

M.S. Computer Science — Machine Learning Specialization | *Stevens Institute of Technology*

Relevant Coursework: Natural Language Processing, Big Data Technologies, Text Mining & Information Retrieval, Machine Learning, Deep Learning, Statistics, Probability Theory, Linear Algebra

B.S. Chemical Engineering | *Rutgers University* *Graduated May 2025*

Coursework: Process Design, Thermodynamics, Mass & Energy Transfer, Calculus, Differential Equations, Material Science

Awards: Cum Laude, Dean's List

LEADERSHIP & ACTIVITIES

Vice President | *ACOBA – East Coast, USA* Jan 2025 – *Present*

- Manage chapter finances and support executive decision-making as Vice President
- Co-organized the largest East Coast cricket festival, overseeing logistics, community partnerships, and charity fundraising demonstrating project management and stakeholder coordination at scale

Member, AIChE Chem-E-Car Team | *Rutgers University* *2024 – 2025*

- Participated in designing, building, and competing with chemically powered model cars in national AIChE competitions, applying engineering problem-solving in a high-stakes team environment

Member, Phi Theta Kappa Honor Society |*Academic Excellence Recognition* *2020 – 2022*

- Inducted for academic excellence and demonstrated leadership; PTK recognizes the top scholars across two-year institutions nationally